



Module Title

Cybersecurity and Networking

Internship Assignment Submission

Ports, Protocols & Services Fundamentals

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1.THEORY

1.1 Answer all 20 questions in your own words

1.What is a network port?

A network port is logical endpoint used by an application or service to communicate over a network.

2. How many port numbers are available conceptually?

There are 65,536 port numbers, from 0 to 65535.

3. What is a network protocol?

A protocol is set of rules that devices use to communicate with each other.

4. What is TCP?

TCP is a reliable, connection-oriented protocol that delivers data in correct order.

5. What is UDP?

UDP is a connectionless protocol with low overhead and no built-in guarantee that data will arrive.

6. Explain two differences between TCP and UDP.

TCP is reliable and uses acknowledgements; UDP does not provide built-in delivery guarantees.

TCP is connection-oriented; UDP is connectionless.

7. What is a listening port?

A listening port is a port where a service is waiting to receive incoming connections.

8. What is an ephemeral/client port?

It is a temporary port usually used by a client to communicate with a server.

9. What is a socket?

A socket is a communication endpoint that combines an IP address, protocol, and port.

10. Which port is commonly used by SSH?

Port 22 (TCP).

11. Which port is commonly used by HTTP?

Port 80 (TCP).

12. Which port is commonly used by HTTPS?

Port 443 (TCP).

13. Which port is used by DNS?

Port 53, commonly using UDP, but TCP is also used in some areas.

14. Which UDP ports are commonly used by DHCP client/server?

UDP 67 for DHCP server and 68 for the DHCP client.

15. What is TCP port 445 commonly associated with?

SMB (Server Management Block), COMMONLY USED FOR Windows file and printer sharing.

16. What is port 3389 commonly associated with?

RDP (Remote Desktop Protocol)

17. What does ss -tuln display?

It displays listening TCP and UDP ports and the addresses they are bound to, using numeric values.

18. Does an open port automatically mean a vulnerability? Explain.

No. An open port only means a service is accepting connections. The service, configuration, software version, and security controls determine whether there is a vulnerability.

19. Why do firewalls use protocol and port information?

Firewalls use them to detect which network traffic should be allowed or blocked.

20. Why are ports and protocols important in cybersecurity?

They help security teams identify services, understand network exposure, configure firewalls, and detect unexpected or unauthorized services.

2. PRACTICAL LAB

TASK 1 — `ss -tuln`: identify TCP listeners, UDP listeners, port numbers and local addresses. Explain `-t -u -l -n`.

```
(kali@kali)-[/home/kali]
PS>
(kali@kali)-[/home/kali]
PS> ss -tuln
```

Netid	State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port
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TASK 3 — `ss -ltn`: record up to THREE TCP listening ports (Port / Address / State). No specific ports required.

```
(kali@kali)-[/home/kali]
PS> ss -ltn
```

State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port
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TASK 7 — `curl -I https://example.com`: note HTTP status, the destination service (HTTPS) and its default port (443). A plain standard request — NOT scanning.